Haixi Zhang

Machine Learning Engineer | Computer Vision | Robotics

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EDUCATION

Cornell University, Ithaca, NY | *M.Eng., Electrical & Computer Engineering* | GPA: 3.81 **Aug. 2024 – May 2025**

University of Rochester, Rochester, NY | *B.S., Electrical & Computer Engineering* | GPA: 3.86 **Sep. 2020 – May 2024**

PROFESSIONAL EXPERIENCE

Machine Learning Engineer, Tensor Auto | Technical Owner: 3 ML/CV Systems San Jose, CA

3D Vision, Multimodal Learning, Python/C++, PyTorch/DDP, VLMs, ONNX/TensorRT, Airflow/Kubernetes, S3 *Sep. 2025 – Present*

- Designed and trained a **6-DoF camera–LiDAR calibration network** with ResNet-18 RGB/GroupNorm LiDAR encoders, **camera-conditioned FiLM**, and **cross-modal attention**, using $SO(3)$ and 3D point-consistency losses; reduced rotation error by **95% (2.95° → 0.14°)** and reprojection error by **89% (22.4 → 2.4 px)** on an **unseen vehicle**.
- Built a **LiDAR–LiDAR calibration network from scratch** using shared GroupNorm spherical-image encoders, **circular azimuth correlation**, and sensor-conditioned rotation/multi-scale translation heads; reduced mean rotation error by **83% (4.86° → 0.83°)** in testing and **63% (0.93° → 0.35°)** in fleet evaluation.
- Built a **two-region Airflow/Kubernetes** pipeline that streamed **17+ TB of raw drive logs from Amazon S3** in a single pass, producing **∼600 GB of camera–LiDAR and LiDAR–LiDAR calibration data**, with sensor-clock drift tracking, pose-aware view selection, frozen rig labels, and per-log retries; shrank the runtime image **20 GB → 600 MB**.
- Accelerated camera–LiDAR training from **∼60 hours to <5 hours (∼13×)** through geometry precomputation, TAR-sharded streaming, and **PyTorch DDP/NCCL**; implemented deterministic sharding, duplicate-free globally aggregated validation, and full-state checkpoint recovery.
- Deployed both calibration models via **PyTorch → ONNX → TensorRT → C++**, with numerical-parity checks, preprocessing alignment, quaternion-aware aggregation, and MAD outlier rejection; camera–LiDAR fleet evaluation showed **44%/57%** lower rotation/translation errors and **50%** lower reprojection error versus nominal extrinsics.
- Fine-tuned **RTMO-s for front- and rear-seat person detection and keypoints** using targeted cabin annotations and COCO data; increased end-to-end recall for held-out forward-leaning rear-seat occupants from **72.1% to 98.6%**, with no increase in empty-seat false positives.
- Designed and delivered **front- and rear-seat occupant monitoring**, integrating person detection, **ResNet-18 occupancy/posture classification** trained with masked multi-task losses, iToF seating-position reasoning, and **VLM temporal fusion** with occupant-change resets and confidence-aware vehicle messaging; system benchmarks reached **99.3% occupancy F1** and **95.8% posture Macro-F1** in real-vehicle testing.

Undergraduate Research Assistant University of Rochester, ECE Department

Python, PyTorch, Graph Learning, HPC, Docker/Kubernetes *Jan. 2023 – May 2024*

- Built reproducible PyTorch/HPC pipelines for graph learning and nature-inspired models in financial and power-system applications, improving benchmark performance by up to **25%**; containerized distributed experiments with Docker/Kubernetes.

TECHNICAL SKILLS

Languages & Systems: C++, Python, CUDA, Bash, Linux, ROS/ROS2, Git, Bazel, CMake
ML & Perception: PyTorch, CNNs/Transformers, cross-attention, multimodal learning, detector fine-tuning, VLM APIs, temporal fusion, RTMO/MMPose
3D Vision & Robotics: Extrinsic calibration, $SE(3)/SO(3)$, camera models, point clouds, OpenCV, Eigen, PCL/Open3D, EKF, SLAM
ML Systems & Data: DDP/NCCL, ONNX, TensorRT, Docker, Kubernetes, Airflow, Amazon S3, sensor synchronization, numerical-parity testing

SELECTED ENGINEERING PROJECTS

Robotic Tool Handler University of Rochester

Lead Developer | C++, ROS, NVIDIA Jetson | GitHub *Jan. 2024 – Jun. 2024*

- Led perception and integration for a seven-member robotics team; deployed camera–LiDAR perception on NVIDIA Jetson with **sub-20 ms inference**, containerized ROS services, and GitLab CI with software-/hardware-in-the-loop validation.

Robot Mapping, Estimation & Navigation University of Rochester

Solo Developer | ROS, C++, LiDAR, TurtleBot2 | GitHub *Jan. 2024 – May 2024*

- Built a C++/ROS navigation stack integrating RGB-D, LiDAR, and wheel odometry for **EKF state estimation**, occupancy mapping, and A* planning; reduced localization error **10%** and planning time **15%**.